

Automatically Identifying Motivation From Text A Public Benchmark and a Large Language Model-Based Scoring System of Intrinsic Motivation in Natural Language

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Abstract—Intrinsic motivation, whether it is the source of a disorder or a necessity to walk through a therapist’s door and carry on with therapy, is central to mental health and therapeutic success. However, tools for assessing motivation remain limited and are largely restricted to closed-ended questionnaires such as the Intrinsic Motivation Inventory (IMI), which can be burdensome to complete and can constrain and bias patients toward a particular framing of intrinsic motivation. This paper seeks to demonstrate that, instead, levels of motivation can be reliably assessed from natural language expressions of motivation using Large Language Models (LLMs), thus allowing for more human-friendly and less time-consuming access to the therapy-critical variable that is intrinsic motivation. This paper introduces two main contributions. First, a public benchmark pairing open-ended natural language expressions of motivation with self-reported IMI scores for six subscales using the standard IMI form, collected from 245 adults, which we call IMIT (IMI score and Text). Second, twelve variants of a prompt-engineered LLM scoring framework that tests a variety of models and trade-offs, including best performance, low cost, and high data security, as well as the quantity of available input data, which we call SIMON (Scoring Intrinsic MOTivation inventory scores based on Natural language). Validation on IMIT shows that the best variant achieves mean absolute error below 15.7% in the worst case. Smaller and locally hosted models incur only limited degradation relative to the best performing model ($\leq 10\%$), supporting privacy-preserving deployment. The paper thus demonstrates the technical feasibility of using LLMs to assess levels of intrinsic motivation, and that these models can meet the privacy and data protection required in therapeutic settings.

Index Terms—Motivation, psychometrics, automated scoring, large language models, natural language processing, affective computing

I. INTRODUCTION

Intrinsic motivation, defined as *the doing of an activity for its inherent satisfaction rather than for some separable consequence* [1], is an essential aspect of mental health and therapeutic activity. Dysfunction in intrinsic motivation is among the most common reasons people seek therapeutic support. It can be a direct symptom and a key mechanism in a variety of mental health disorders, manifesting as deficiency (e.g., depression [2]–[4]) or instability (e.g., bipolar disorders [5], [6]). Intrinsic motivation (IM) can also act as a precursor and risk factor for other disorders (e.g., high intrinsic motivation regarding work as a risk factor for burnout, low intrinsic

motivation as a risk for academic and professional failure [7]–[9]). Additionally, intrinsic motivation is a core determinant of therapeutic success, including deciding to initiate therapy and acting on that decision [10]–[12], mobilizing resources to engage in therapeutic activities and optimizing effectiveness [13], [14], and avoiding dropouts [15]–[17]. A substantial part of therapeutic work involves, directly or indirectly, creating and maintaining IM, including developing a strong therapeutic alliance [18]–[20], establishing therapeutic goals and plans [21], [22], engaging in exercises such as value clarification and meaning-making [23]–[25], and supporting patients’ autonomy [16], [26], [27]. Therapeutic work therefore requires the capacity to assess and manage patients’ intrinsic motivation as a clinical resource, both directly (e.g., anticipating whether a patient will be receptive or reluctant toward specific approaches) and indirectly by identifying factors that may impact intrinsic motivation (e.g., addressing sleep deprivation to increase the availability of IM) [28]–[30]. Failing to account for intrinsic motivation may have negative consequences, such as a dramatic increase in suicidality when depressed patients regain intrinsic motivation [31]–[33].

However, despite the importance of intrinsic motivation for therapeutic success, current methods for assessing intrinsic motivation remain limited. Intrinsic motivation is commonly assessed informally through therapists’ empathic judgment [34]–[36], which can make assessment depend on individual therapist sensitivity, both in general and with the specific patients they work with. A second approach relies on standardized forms, such as the *Intrinsic Motivation Inventory* (IMI) [37]–[39], which derive intrinsic motivation scores from self-report response scales. While such instruments provide quantifiable and standardized measures, they also share typical drawbacks of close-ended questionnaires [40]–[42]. They impose a framing of intrinsic motivation that may prime responses and broader conceptions of intrinsic motivation [43]–[45]; they may systematically exclude certain experiences or populations (e.g., neurodivergent or non-native respondents) [46]–[48]; they may be too long or intimidating for patients experiencing psychological distress [49]–[51]; they inherently limit the collection of qualitative insights (e.g., *what* motivates the respondent) [52]–[54]; and they introduce an interaction format that can feel bureaucratic within therapeutic processes that depend on empathy and care.

The current paper seeks to expand the state of the art by demonstrating the viability of acquiring reliable quantitative assessments of intrinsic motivation from patients’ natural lan-

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guage self-reports. This alternative assessment method offers practical benefits for making future mental healthcare more scalable, personalized, and accessible. With adequate privacy measures, natural language input can support population screening at scale, early detection of IM-related disorders, and thus cost-effective interventions [55]–[57]. Natural language-based assessments can replace or complement classic intrinsic motivation forms during onboarding, providing therapists not only an intrinsic motivation level estimate but also insights into which motivational aspects may be impacted [58]–[60]. Because it relies on a convenient natural language format, it can also be integrated throughout therapy (e.g., weekly prompts in which patients write or speak a few sentences about their motivation), enabling fine-grained, personalized tracking of intrinsic motivation over time [61]–[63]. From a research standpoint, natural language-based assessments allow for exploring new fields of motivation for therapy research by making accessible a wide range of already existing data sources (e.g., social media, forums, therapeutic journaling, patient-therapist chats in Internet-Based Cognitive Behavioral Therapy) [64]–[66], as well as allowing for facilitated future data collection. Reliable intrinsic motivation assessment from natural language self-reports can also function as an enabling technology for scalable approaches to mental health with greater (e.g., economic, geographic, social) accessibility, including Internet-Based Cognitive Behavioral Therapy and AI-powered automation such as therapeutic chatbots [62], [67], [68]. In such settings, natural language-based assessments can provide personalized qualitative insights into factors shaping intrinsic motivation that remote clinicians or AI systems can use to tailor support [69]–[71]. Specifically, this paper addresses the following research questions:

- **Research Question 1:** Can quantitative measures of intrinsic motivation captured by Intrinsic Motivation Inventory (IMI) scores be retrieved from natural language expressions of motivation using Large Language Model (LLM)-based scoring?
- **Research Question 2:** Are scores produced by LLM-based scoring reliable in absolute terms, or do they need to be transformed to recover actual IMI scores?
- **Research Question 3:** How does using smaller models and/or in-house LLMs (enabling additional data control and privacy protection) impact the performance of LLM-based IMI scoring?
- **Research Question 4:** How do different methods for eliciting natural language expressions of intrinsic motivation impact the performance of LLM-based IMI scoring?

As main contributions and methods, this paper addresses these research questions as follows. First, we developed a series of LLM-based scoring systems to estimate IMI scores from natural language expressions of IM. We call this system SIMON (Scoring Intrinsic MOTivation inventory scores based on Natural language), implemented via prompt engineering. While sharing the same prompt structure, SIMON variants implement these prompts across LLMs with different trade-offs in expected performance, cost, and capacity for stronger data and privacy protection: gpt-5.2, gpt-4o, gpt-5-nano,

and softaibox-openaioss-120b. We refer to these as SIMON-gpt-5.2, SIMON-gpt-4o, SIMON-gpt-5-nano, and SIMON-softaibox-openaioss-120b. Second, to obtain ground-truth data, we developed and deployed a form that pairs respondents’ natural language expressions of intrinsic motivation about their primary activity with quantitative intrinsic motivation scores from standard IMI forms [72]–[74]. We call the resulting database Intrinsic Motivation Inventory score and Text (IMIT). While SIMON is intended for clinical research, IMIT focuses on primary activities to validate SIMON’s approach before collecting sensitive clinical data and to enable IMIT to be released as a public benchmark, an essential step for long-term research on identifying markers of motivation. Finally, we validated SIMON by assessing whether SIMON scores computed from IMIT participants’ natural language responses match or correlate with the participants’ self-reported IMI scores. Participants’ language responses were analyzed either in full or in truncated form to evaluate how different elicitation methods affect the ability to recover IMI scores from natural language.

II. PREVIOUS WORK

To the best of our knowledge, this paper is among the first to use Large Language Models to derive intrinsic motivation scores from text and to compile a dataset that links natural language expressions of motivation with intrinsic motivation scores. The closest related work is a study that adapted a pre-existing dictionary-based method (called Linguistic Inquiry and Word Count) [75] to assess motivation based on textual content [76]. While our goals are related, the soundness of this method for scoring intrinsic motivation from natural language content has not been validated. Some partial reliability metrics were reported in [75], suggesting that this approach had low internal reliability.

Broadening the scope, a small yet rapidly growing body of research examines Artificial Intelligence, Natural Language Processing, and Large Language Model methods for mapping psychometric scores onto natural language text [77]–[79]. This literature includes approaches such as machine-learning methods based on lexical features (e.g., counting the frequency of words from a predefined dictionary) [80], machine-learning methods based on embedding representations of text using Bidirectional Encoder Representations from Transformers (BERT) [81], [82], and direct classification and scoring of natural language input using Large Language Models [83]. These tools have been applied across domains, including general psychology for identifying personality [80] and education for identifying types of reasoning [84]. Focusing on clinical psychology and psychiatry [85], applications of these methods have been reported, including for scoring levels of anxiety [81], depression [86]–[89], stress [90], [91], psychiatric functioning [92], and patient engagement [93].

III. THEORETICAL BACKGROUND: INTRINSIC MOTIVATION INVENTORY

The Intrinsic Motivation Inventory (IMI) is a multidimensional self-report instrument designed to assess participants’

subjective experience during a specific target activity [94]. The IMI has received extensive empirical validation [94], [95], [95]–[102]. The IMI has been repeatedly deployed in experimental and applied research as a practical index of intrinsic motivation, including clinical psychology [103]–[105], wellbeing [106], psychology and psychiatry training [107], [108], sport [94], [95], [95], school [99], [109], gaming and human-computer interaction [110], as well as across a variety of languages and cultural settings [106], [109], [111].

The IMI originates directly from the Self-Determination Theory [101], [112], specifically the Cognitive Evaluation Theory [96]. The IMI frames intrinsic motivation as the psychological mechanisms driving individuals to engage in an activity for the inherent expected satisfaction tied to performing the activity rather than for external consequences (e.g., material or social benefits). At its core, the IMI introduces a set of indicators that are expected to be relevant for identifying factors of intrinsic motivation, whether positive (e.g., experiencing enjoyment when engaged with the activity) or negative (e.g., experiencing anxiety while doing the activity). These include the presence of supportive social-contextual conditions that nurture autonomy and competence, and the absence of external contingencies (e.g., control, pressures). Within the current standardized framework, these indicators are organized in the following six subscales:

- 1) interest/enjoyment
- 2) perceived competence
- 3) effort/importance
- 4) pressure/tension
- 5) perceived choice
- 6) value/usefulness

Interest/Enjoyment covers the extent to which individuals experience intrinsic pleasure, curiosity, and engagement during the activity. Perceived Competence covers self-evaluations of capability and effectiveness in performing the activity, reflecting confidence in achieving task demands and desired outcomes. Effort/Importance covers the degree of exertion and investment individuals report allocating to the activity, as well as the subjective importance attributed to performing well or persisting. Pressure/Tension covers the level of experienced stress, apprehension, or performance-related strain during the activity. It reflects a negative affective state that is often inversely associated with intrinsic motivation. Perceived Choice covers the extent to which engagement in the activity is experienced as volitional and self-endorsed rather than externally compelled, reflecting perceived autonomy in participation. Value/Usefulness covers the perceived personal relevance, utility, or instrumental value of the activity for attaining meaningful goals or outcomes, reflecting internalized reasons for engagement.

The IMI is operationalized as a questionnaire available online¹ that lists personal statements relative to indicators of motivation (e.g., “I enjoy pursuing this activity very much”) and requires respondents to assess how much they endorse these statements via 7-point Likert-type responses, that is, scales ranging from one to seven, where 1 indicates low

endorsement and 7 indicates high endorsement. The most complete questionnaire covers all six subscales and includes 45 items, with four to seven items per subscale. The overall IMI scoring consists of a six-dimensional score, where each subscale corresponds to a dimension. For each subscale, the score consists of the average of the Likert responses, with negative indicators reversed (e.g., a score of two for the negative indicator “I think this is a boring activity” is associated with a score of $8 - 2 = 6$ to be factored into the average for the interest/enjoyment subscale).

IV. METHOD

A. Overview

In the absence of a cohesive benchmark linking IMI scores with open-ended expressions of intrinsic motivation in natural language, we first needed to collect a database, which we call IMIT (for IMI scores and Text). IMIT connects, on the one hand, open-ended qualitative free-text questions about the respondents’ intrinsic motivation (seven in total, one for each IMI subscale, plus a general motivation text) and, on the other hand, close-ended quantitative one-to-seven scale questions on self-reported levels of motivation based on the standard IMI form. A survey was developed and deployed to collect these data, resulting in the IMIT database. IMIT serves as gold-standard data connecting numeric IMI scores and open-ended writing about motivation.

Second, to show that IMI scores can be derived from natural language answers to motivation-related questions using LLMs, we developed a series of twelve LLM-based classifiers, which we call SIMON (for Scoring Intrinsic MOTivation inventory scores based on Natural language), using prompt-engineering methods. SIMON associates an IMI score for each of the six IMI subscales based on natural language answers to motivation-related questions. The variants of SIMON were selected to allow a comparison across a diversity of ecologically valid use cases, including: a) *controlled research environments*, where maximal scoring performance is the main concern and responsiveness, costs, and data security are secondary (e.g., because data are anonymized or respondents have provided consent); b) *business environments*, where scoring performance may be traded off for cost-efficiency and responsiveness (e.g., a mobile app, where users expect answers within a few seconds); and c) *sensitive contexts*, where high privacy and data security are hard constraints, followed by scoring performance (e.g., as part of an intake procedure in clinical psychology). Additionally, we compared how the amount and type of data that can be acquired from respondents impact performance, as various contexts may allow more or less access to data. SIMON is then validated against the gold-standard data from IMIT by assessing whether the numeric scores inferred by SIMON, based on the natural language responses provided in IMIT, align with the numeric values provided by the human respondents in IMIT.

B. IMIT Data acquisition

1) *Survey Development*: Data acquisition consisted of developing and disseminating a survey involving both natural

¹<https://selfdeterminationtheory.org/IMI-inventory/>

language depictions of motivation and the standard IMI scale form. To enable participation from a broad range of respondents without collecting sensitive data, the survey assessed intrinsic motivation regarding participants' primary activity.

The form was split into two parts. In the first part, the form included a general open-ended question about motivation (i.e., “How would you describe your motivation for your primary activity?”), followed by one open-ended question for each IMI subscale, based on the description of that subscale and on questions prescribed by IMI theory and the standardized form (e.g., “With regards to feelings of pressure/tension and anxiety/relaxation, how do you generally feel during this primary activity?” was asked for the “pressure/tension” subscale). In the second part, for each subscale, we used the questions included in the related standard IMI form specified for that subscale, which consists of a series of four to seven Likert-scale questions (e.g., for the “pressure/tension” subscale, participants were asked, as part of a set of questions, how true the following statement is for them: “I feel very tense while doing this activity.”, from 1 (not true at all) to 7 (very true)) [97]. This two-phase approach was selected to have participants answer the natural language questions before being exposed to the conceptual framing embedded in close-ended questionnaires. This approach is intended to be straightforward to transfer to clinical contexts by replacing the focus on the primary activity with, for example, general motivation, motivation regarding therapy, and/or motivation regarding specific parts of therapy.

2) *Data collection*: The form was published through Connect, an online source of high-quality participants by CloudResearch (formerly TurkPrime) [113]. Respondents were offered a \$3 compensation for completing the survey, which was expected to take approximately 15 minutes. The Prime Panel function of CloudResearch was used to gather a balanced sample of respondents in terms of age, gender, and educational background.

The final IMIT database contains $N = 245$ responses. A total of $n = 262$ adults responded to the questionnaire, of which $n = 12$ responses were removed due to a missing primary activity and $n = 5$ responses were removed due to incompleteness.

3) *Ethical considerations*: All procedures in this study complied with the ethical guidelines of the Swedish Research Council and the principles of the Declaration of Helsinki. Participants provided informed consent before completing the survey and were informed about the purpose of the study, the voluntary nature of participation, their right to withdraw at any time, and the public availability of anonymized data. The consent form specified that no identifying information (e.g., name, address) should be provided and that any inadvertently provided personal details would be removed before publication. The data are available here².

C. SIMON Automated Scoring System Development

A series of twelve variants of SIMON were produced to assess whether IMI scores can be retrieved from texts eliciting

motivation statements from IMIT across a variety of ecological use cases. To cover these use cases, two variables were controlled: model type and amount of input.

The *model type* variable had four possible values, namely *gpt-5.2* and *gpt-4o*, representing computationally extensive LLMs with the highest expected scoring performance³; *gpt-5-nano*, representing cheaper and more responsive models; and *softaibox-openaioss-120b*, representing locally hosted LLMs offering strong data protection features (e.g., GDPR-compliance). The *input data* variable had three possible values, namely *minimalist* (i.e., the answer to the general question about motivation), *subscale-based* (i.e., if subscale s is assessed, the answer to the natural language question relative to s), and *exhaustive* (i.e., the answer to the general motivation question and to all six natural language questions associated with the six IMI subscales). Each variant of SIMON is named after a combination of these two variables (e.g., SIMON-gpt-5.2-exhaustive). The model type variable allows addressing Research Question 1 (by selecting the top-performing model) and Research Question 3 (by comparing the qualities of the various models). The input data variable allows addressing Research Question 4.

1) *Prompt Engineering*: Prompt engineering methods were applied to maximize the expected scoring performance of the various SIMON implementations, based on the frameworks provided by [114]–[116]. These frameworks involve the following steps: 1) assign a role to the LLM (i.e., provide a role for the LLM to adopt, together with relevant contextual elements); 2) specify a task to be completed (i.e., describe in detail the task to be performed, along with precise instructions for executing it), including 2.a) requiring a justification for the choice to be made (commonly referred to as a chain-of-thought prompt engineering method [117], i.e., asking the LLM to first justify which elements in the text correspond to the code to be assigned [118]), 2.b) requiring a decision, and 2.c) defining an output format that can be directly computationally processed (JSON in our case); and finally 3) provide the content to be assessed.

A set of prompts was generated, with one prompt per subscale and input type. Changing the subscale was performed by changing the task assigned to the system. The input was adapted as follows: for the exhaustive approach, all questions asked and answers provided by the respondent were appended to the prompt; for the subscale-based approach, only the question and answer relative to the assessed subscale were appended to the prompt; for the minimalist approach, only the general question and related answer were appended to the prompt. To further improve expected performance, we considered including additional content about each subscale from IMI-related research, but did not proceed to avoid infringing IMI licensing⁴. The exact prompt structure is provided in Annex A.

2) *Model specification*: The various models were accessed through either the OpenAI API for *gpt-5.2*, *gpt-4o*, and *gpt-5-nano*, or our in-house LLM API for *softaibox-openaioss-120b*.

³Both *gpt-5.2* and *gpt-4o* were used as they are commonly argued to be comparable in terms of performance

⁴<https://selfdeterminationtheory.org/IM-inventory/>

²The data will be linked here to the publication through the IEEE data publication system.

The following parameter settings were selected: temperature = 0.0 and top_p = 1.0 for gpt-4o and softaibox-openaioss-120b. Temperature and top_p are not available parameters for gpt-5.2 and gpt-5-nano.

D. Statistical Analyses

The statistical analyses were conducted to assess whether the IMI scores estimated by SIMON from the textual data can be relevant for predicting the IMI scores reported by the participants, whether by directly using the score provided by the LLM (using the Mean Absolute Error, MAE) or indirectly (using Spearman’s rank correlation coefficient, ρ). Additionally, confusion matrices were used to identify possible systematic biases in SIMON scoring. The MAE allows addressing all research questions, and the correlations and confusion matrices allow further substantiating Research Question 2.

All statistical analyses were conducted in Python (version 3.11.5), utilizing SciPy.stats, Statsmodels, NumPy, and matplotlib libraries for data preprocessing, visualization, and inferential statistics.

1) *Mean Absolute Error*: The Mean Absolute Error (MAE) quantifies, given a dataset of pairs of numbers, the average distance between these numbers. Formally, the MAE of a dataset is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

where n is the size of the dataset, and y_i and $\hat{y}_i$ represent the i^{th} pair of numbers in the dataset. MAE was used to assess the average distance between SIMON IMI scores inferred from text and gold-standard IMI scores, which are scaled between 1 and 7.

MAE provides a straightforward measure for assessing the scoring performance of SIMON on various IMI subscales by summing the distances between the IMI score directly reported by participants on that subscale and the IMI score retrieved by SIMON based on the participants’ open-ended natural language answer. Note that the F1 score was not used, as this metric is tailored for binary classification tasks, which differ from the scoring task investigated in this paper. MAE is better suited because it captures the mean deviation of predicted scores from observed scores without penalizing variance directionality (Vujovic, 2021).

2) *Spearman’s Correlation*: Spearman’s correlation (Spearman’s ρ) is a standard nonparametric measure of association that quantifies the strength and direction of a monotonic relationship between two variables based on the ranks of their values. Spearman’s correlations are measured by a score ρ , which represents the strength and direction of the correlation. Conventionally, $0 < \rho < .10$ means a small positive correlation, $.10 \leq \rho < .30$ means a medium positive correlation, and $\rho \geq .50$ means a large positive correlation. Likewise, $-.10 < \rho < 0$ means a small negative correlation, $-.30 \leq \rho < -.10$ means a medium negative correlation, and $\rho \leq -.50$ means a large negative correlation, as per the conventions of [119].

Additionally, the statistical significance of this correlation was assessed using a two-sided hypothesis test of ($H_0 : \rho = 0$). The associated value, commonly referred to as the p -value, represents the probability, under the null hypothesis of no monotonic association, of observing a Spearman correlation at least as extreme as the one obtained in the sample (in absolute value). Small p values indicate that the observed association is unlikely to be due to random sampling variation alone, given (H_0). p -values were adjusted for multiple comparisons using the Holm-Bonferroni method to control the family-wise error rate.

The uncertainty in ρ was measured using the standard $(100(1 - \alpha)\%)$ Confidence Interval (CI). The CI provides a range of plausible values for the population Spearman correlation that is compatible with the observed data under the assumed sampling process. Confidence intervals were computed using bootstrapping, which resampled the paired data 5000 times, recalculating ρ each time. The 2.5th and 97.5th percentiles were used to create lower and upper bounds of the bootstrapped Spearman distribution. These confidence intervals are reported alongside point estimates to convey both effect size and precision.

Spearman’s correlations were applied to establish, for each IMI subscale, the correlations between the IMI scores directly provided by participants and the IMI scores inferred from the natural language input provided by these participants. These correlations were computed on the top-performing variant of SIMON (i.e., SIMON-gpt-5.2-exhaustive). This metric allows answering Research Question 2, as correlations can indicate whether corrective transformations can be applied to the IMI score produced by SIMON to better align with the actual IMI score. For example, if SIMON were to overestimate all actual IMI scores by three points, the MAE would be 3 whereas the correlation would be perfect ($\rho = 1$). A high correlation signifies that the original IMI score can be recovered from the one retrieved by SIMON after transformations.

3) *Confusion Matrices*: We evaluated scoring behavior using confusion matrices, i.e., contingency tables that cross-tabulate ground-truth IMI scores provided by respondents (rows) against scores estimated by SIMON (columns). These matrices decompose the decisions into counts of correct and incorrect decisions [120]–[122]. To discretize the IMI scores provided by respondents, which are real numbers computed as averages, these scores were rounded to the nearest integer.

The resulting matrices are 7x7 matrices, one matrix per subscale, where the diagonal cells represent correct estimations and other cells represent erroneous classifications. The farther a cell is from the diagonal, the larger the error [121], [122]. An analysis along a given row allows assessing how accurate SIMON’s assessments are for a given subscale, given a target score provided by participants on that subscale. An analysis along a given column allows assessing whether a given score assigned by SIMON is reliable, or whether it is regularly used to qualify another score actually provided by participants. A matrix-wide analysis allows identifying systematic under- or overestimation, if cells below (respectively, above) the diagonal are associated with higher counts than cells above (respectively, below) the diagonal.

This analysis was applied on the top-performing variant of SIMON (i.e., SIMON-gpt-5.2-exhaustive). This analysis allows substantiating Research Question 2 on systematic errors that the raw output of SIMON may produce.

E. Qualitative analysis

A qualitative analysis was conducted on $n = 10$ answers for the *exhaustive* input data for all six subscales, as well as on the associated justifications produced by SIMON, to substantiate all research questions with practical insights from the data. This step addresses Research Question 1 by allowing us to identify whether scoring errors arise from the data or from LLM limitations, as well as to qualify content-specific variables that can explain how the system performs (e.g., very short or very long responses may confuse the system).

In a first step, human labelling was applied to the various entries to assess whether the natural language answers provide adequate epistemic material for IMI scores to be reliably retrieved. This step allows distinguishing whether prediction errors arise from technical limitations of SIMON’s approach or whether prediction errors are inherent to imprecision in the data (e.g., variability across respondents, noise in textual expressions of motivation) [123]–[125]. This distinction can be examined by assessing whether the human labeller and SIMON associate different scores to the same text, and whether two texts from two participants that are arguably similar are associated with different respondent-provided IMI scores [126].

In a second step, the relevance of the justifications produced by SIMON was assessed against the input text on which they are based, and against the numeric values assigned by SIMON. While the justifications used by LLMs do not constitute scoring grids that the LLM scrupulously relies on for scoring the text [127]–[129], a systematic mismatch between the text and the justification, or between the justification and the final score, can be indicative of underlying limitations in the capacity of the LLM to process this data [130]–[132].

V. RESULTS

A. Descriptive statistics

The final number of participants retain for analysis was $N = 246$. The average age was $M_{age} = 38.9$, $SD = 10.70$ and the gender ratio was 56% men, 41% women, 2% non-binary/other. Participants’ primary activities were work for 88% of the participants and study for the remaining 12%.

B. Comparing Human-Provided IMI Score Against IMI Scores Inferred by SIMON

1) *Mean Absolute Error of SIMON’s Estimates:* The MAE of each variant of SIMON per subscale, i.e., the average distance between the IMI scores inferred by SIMON from open-ended natural language answers and the IMI scores directly provided by the participants, is presented in Figure 1. The higher the MAE, the larger the average error.

As an overall observation, leaving aside the minimalist input data, the MAE remains within a range between 0.5 and 2 for

all subscales and variants of SIMON. When considering the best-performing variants of SIMON per subscale, all subscales have an MAE below 1.1, with half of the subscales below 0.8. This indicates that high-performing solutions are relatively close to participants’ self-reports on a 1-to-7 scale. The most computationally expensive model (i.e., SIMON-gpt-5.2-exhaustive) performs best or second best on all subscales and outperforms SIMON-gpt-4o on all subscales. The model selected for lower cost and higher responsiveness (i.e., SIMON-gpt-5-nano-exhaustive) and the model selected for stronger data protection features (i.e., SIMON-openaioss-exhaustive) perform slightly worse than the best-performing variant, while remaining relatively close (within a 10% relative difference). Likewise, using the exhaustive input tends to achieve higher scoring performance than the subscale-based input, although the difference remains limited. Using the minimalist input tends to lead to a substantial degradation in performance compared with the other two.

2) *Correlation Analysis:* Spearman’s rank correlations (ρ) between IMI scores reported by participants and IMI scores inferred by the various implementations of SIMON from participants’ free-text responses are presented in Table I.

When comparing by model, SIMON-gpt-5.2-exhaustive appears to yield, on average, the strongest correlation values between IMI scores provided by participants and IMI scores inferred by SIMON from open-ended natural language responses. However, these correlation values are only marginally higher than those of the other models, which appear to be broadly comparable. When comparing by input data, correlations are highest when the exhaustive input is used. They are, on average, slightly above those obtained with the subscale-based input and substantially above those obtained with the minimalist input, for which correlations are often low except for the Interest/Enjoyment subscale. When comparing by subscale, considering only the exhaustive and subscale-based inputs, Interest/Enjoyment achieves the highest correlations by far. Pressure/Tension, Perceived Choice, and Value/Usefulness also show strong correlations in most cases, followed by Perceived Competence and Effort/Importance, which tend to show medium correlations.

3) *Analysis of Scoring Errors:* The confusion matrices of the top-performing model (i.e., SIMON-gpt-5.2-exhaustive) for each IMI subscale are illustrated in Figure 2. This figure allows identifying whether the model exhibits recurrent errors in its assessments, thus indicating whether corrective measures may be warranted.

When reading each matrix along its diagonal from top left to bottom right, which corresponds to accurate predictions, it can be observed that, in most cases, the most frequent (i.e., lightest) cell in each row is either on the diagonal or directly adjacent to it. Likewise, frequencies tend to drop sharply as cells move farther from the diagonal. Overall, this pattern suggests that SIMON’s allocations align with expectations. While rare, divergences of four points or more between the estimated and actual scores can be observed.

When examining error patterns across rows, SIMON appears to be least accurate when estimating scores for respon-

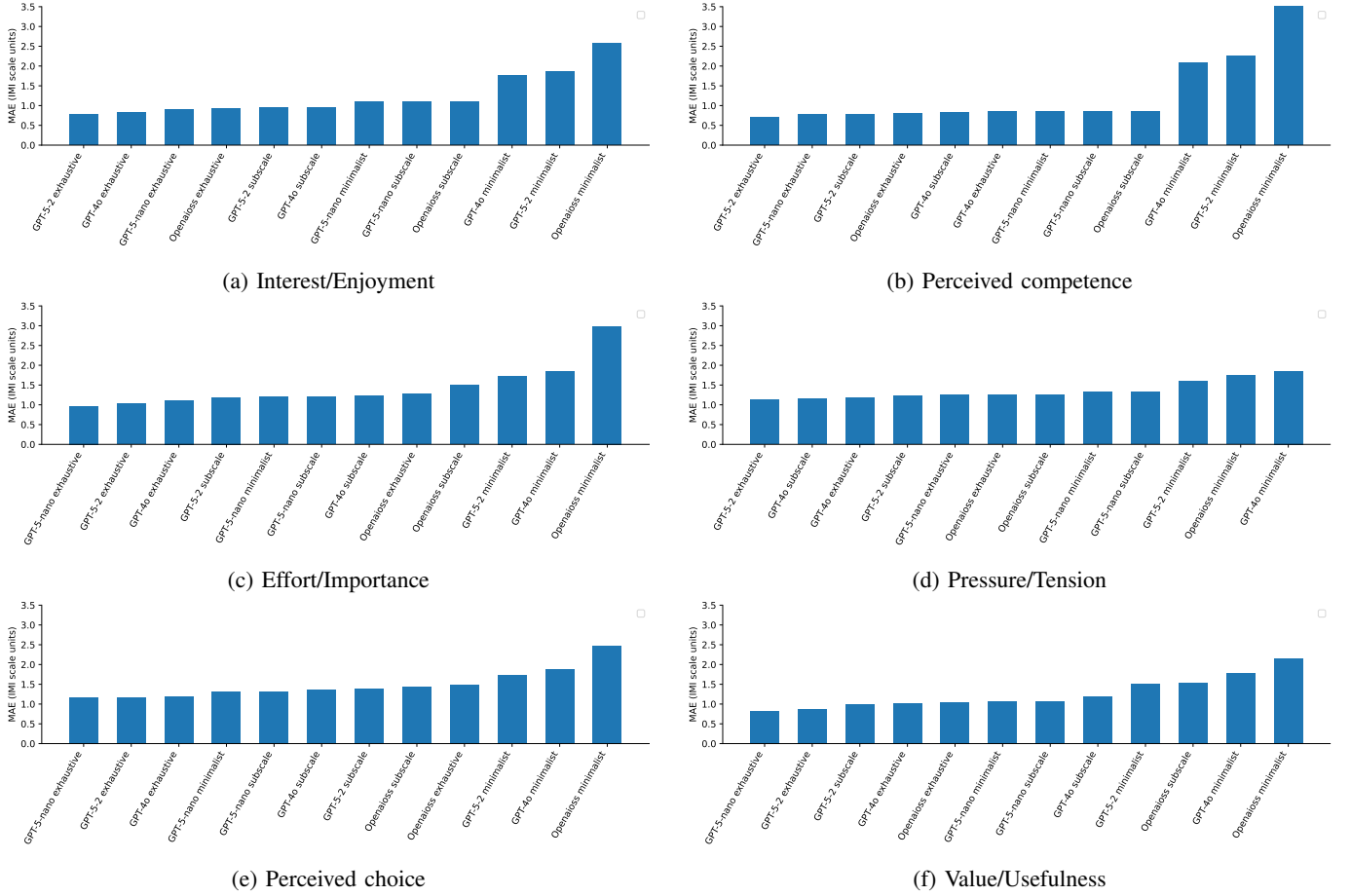


Fig. 1: Means Absolute Error between SIMON’s estimated IMI score and the IMI score provided by the respondents.

TABLE I: Spearman’s ρ correlation score between the Intrinsic Motivation Inventory (IMI) scores provided by the respondents and the IMI scores inferred by SIMON from the respondents’ text for all six IMI subscales and twelve variants of SIMON. “E” means “exhaustive”, “S” means “subscale-based”, and “M” means “minimalist”. ** indicates $p < 0.01$ after Bonferroni-Holm correction.

Model Subscale / Input data	gpt-4o			gpt-5.2			gpt-5-nano			softaibox-openaiooss-120b		
	E	S	M	E	S	M	E	S	M	E	S	M
Interest / Enjoyment	0.77**	0.72**	0.41**	0.79**	0.67**	0.41*	0.74**	0.67**	0.35**	0.75**	0.65**	0.34**
Perceived competence	0.36**	0.36**	-0.04	0.48**	0.42**	0.03	0.34**	0.37**	-0.03	0.40**	0.45**	-0.01
Effort / Importance	0.38**	0.29**	0.23*	0.38**	0.36*	0.13	0.37**	0.29**	0.12	0.43**	0.30**	0.10
Pressure / Tension	0.58**	0.62**	0.08	0.60**	0.60**	0.13	0.60**	0.59**	0.06	0.59**	0.57**	0.03
Perceived choice	0.56**	0.53**	0.16	0.60**	0.50**	0.21**	0.52**	0.47**	0.09	0.57**	0.50**	0.17
Value / Usefulness	0.49**	0.35**	0.21	0.44**	0.32**	0.19**	0.41**	0.27**	0.16	0.45**	0.33**	0.21

dents who provided the maximal score (i.e., 7), with a strong tendency to underestimate these scores as 6. When considering the overall distribution of errors, a slightly larger proportion of errors fall below the diagonal. This indicates a mild bias toward underestimating predicted scores.

C. Qualitative Analysis

1) *Reliability of IMIT Semantic Contents*: Part of the qualitative analysis assesses whether SIMON’s scoring performance reflects SIMON’s capabilities or the semantic content of the collected natural language data. This analysis compares

whether human annotations reproduce SIMON’s scores and whether qualitatively similar texts are associated with different IMI scores provided by participants.

While the number of annotations is insufficient for statistical validity ($n = 10$), human-based scoring of the open-ended texts arguably aligned with the IMI scores provided by SIMON, which were themselves mostly aligned with participants’ self-reported scores. The annotator also identified that, on occasion, significant discrepancies occur across participants despite similar texts. For instance, one participant referred to their activity as “a life of self-affirmation” and emphasized that they “find value in public service,” whereas another participant

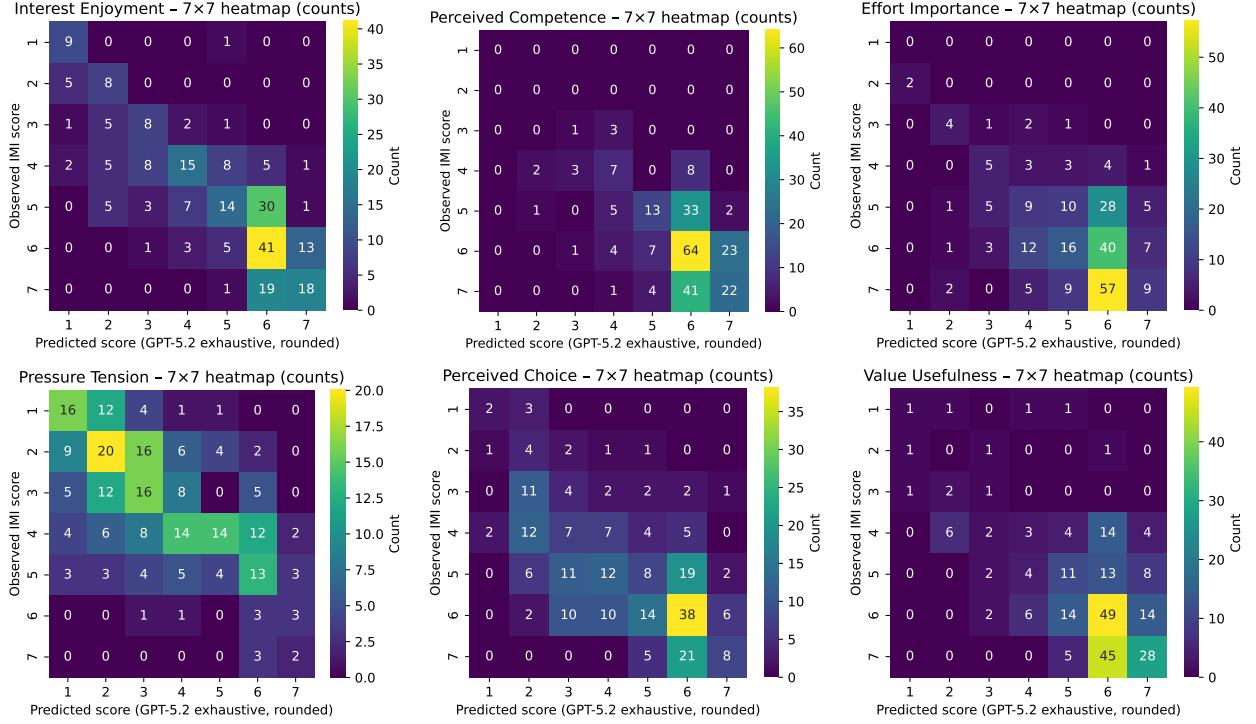


Fig. 2: Confusion matrices for the top-performing variant of SIMON (i.e., SIMON-gpt-5.2-exhaustive) for each IMI subscale. The cell in row i and column j represents the number of times SIMON-gpt-5.2-exhaustive assigned score j when the participant scored i on the IMI form (rounded to the nearest integer). The diagonal where $i = j$ represents correct predictions. The color of each cell represents the number of occurrences.

stated that their work is “very meaningful and important” and helps them “better [themselves] as a person” and “show up better” for loved ones. Although both respondents arguably express a marked sense of meaning and personal growth, their self-reported Value/Usefulness scores differed, from 7.00 for the first participant to 5.25 for the second. Both the human annotator and SIMON assigned similar scores to these two participants. See Annex A for an example of alignment between human and SIMON scoring that diverges from self-reported scores.

2) *Reliability of SIMON’s Reasonings*: The reasoning traces of these $n = 10$ samples were examined to assess the relevance of the LLM’s justifications, considering the quality of the input text, the interpretive strategies produced by SIMON, and the score assigned by SIMON.

Across cases, the model generally grounded its reasoning in explicit quotations from the input text whenever possible. In particular, affective expressions (e.g., “I enjoy”, “I feel stressed about”) and explicit references to perceived capability (e.g., “I think I’m good at”) were retrieved. These explanations were generally aligned with the subscale being assessed. In the absence of explicit markers, the model often extrapolated beyond the explicit content of the text, for example by inferring motivational states from ambiguous or weakly related phrasing (e.g., interpreting brief mentions of routine or workload as indicators of high effort). In other instances, the model understated tension or stress when participants used subtle or indirect language to describe discomfort. The model also

occasionally introduced information that was not present in the participant text. This issue was particularly prevalent when participants described neutral or mixed feelings or relied on implicit cues, in which case the model sometimes introduced unwarranted confidence in their motivation or competence. More extrapolative explanations were associated with lower scoring performance. See Annex A for an example of such an overinterpretation.

VI. DISCUSSION

A. *Research Question 1: Can quantitative measures of intrinsic motivation captured by Intrinsic Motivation Inventory (IMI) scores be retrieved from natural language expressions of motivation using Large Language Model (LLM)-based scoring?*

1) *Insights*: The results show that LLM-based scoring of open-ended motivational text (specifically demonstrated by SIMON-gpt-5.2-exhaustive) can reliably retrieve IMI scores, approximating the actual IMI score within 0.5 to 1 point depending on the subscale. These results are further supported by correlational analyses, which show moderate to strong correlations between the scores produced by SIMON and the IMI scores provided by respondents, depending on the subscale.

The qualitative analysis complements these insights by confirming that, while the IMIT database is generally sound (i.e., open-ended textual data collected in IMIT is usually sufficient

to closely approximate numeric IMI scores), self-reported intrinsic motivation captured by IMI scores occasionally differs across individuals despite their texts being arguably similar in nature. This variability, which is common for psychological constructs, appears relatively limited within IMIT, supporting the relevance of the dataset for linking textual and numeric accounts of motivation.

SIMON tended to perform better, both quantitatively and in the qualitative soundness of explanations, when participants relied on explicit cues of motivation and related affect rather than on implicit or indirect cues. These systematic errors indicate that LLM-based scoring is sensitive to explicit cues.

2) *Significance*: These results demonstrate the viability of using LLM-based solutions for recognizing intrinsic motivation for research and clinical purposes. This technical possibility can complement traditional methods for eliciting intrinsic motivation, including direct interviews, which may be too expensive or impracticable to implement, and form-based measurement, which may be perceived as alienating and normalizing compared with the natural language expression of one’s experience permitted by SIMON [133]–[135]. Indirectly, the technological prospect carried by SIMON enables a wide variety of benefits, such as in-depth tracking of patient motivation over time, whether with respect to motivation-related conditions tied to a disorder (e.g., depression) or with respect to therapeutic engagement (e.g., dropout risks, willingness to engage in difficult exercises) [63], [136], [137]. With an adequate setup, such as processing therapeutic journals generated as a side-product of therapeutic activities, SIMON-like methods may enable cost-free insights on a core variable for mental health and therapeutic success [61], [137], [138].

3) *Limitations and future work*: The collected evidence is limited to the current dataset and the specific implementation of SIMON variants. While the proposed data collection method is a straightforward attempt at eliciting respondents’ intrinsic motivation, alternative elicitation methods may provide data that are even more informative for accurate IMI assessments. Examples include requiring further elaboration from participants or using an automated chatbot that asks relevant follow-up questions [139]–[141]. The participant pool was limited to online respondents willing to fill in forms. Future work can further assess generalizability by repeating this study in different contexts (e.g., different sociocultural groups).

Likewise, alternative methods for analyzing the text (e.g., LLM-based, AI-based, or statistical methods) may further improve performance, for example via fine-tuned models given the availability of a much larger dataset [82], [142], [143]. Additionally, manual scoring of IMIT open-ended texts can expand the qualitative analysis performed here and provide statistically grounded estimates of whether SIMON meets human performance.

B. Research Question 2: Are scores produced by LLM-based scoring reliable in absolute terms, or do they need to be transformed to better approximate actual IMI scores?

1) *Insights*: The MAE results of the best-performing variant of SIMON (SIMON-gpt-5.2-exhaustive) indicate that di-

rect prompting provides an acceptable baseline. The correlation analysis complements this finding by identifying medium to strong correlations between IMI scores provided by participants and IMI scores estimated from participants’ open-ended text. This suggests that higher precision may be achievable through corrective transformations applied to raw SIMON outputs. In particular, systematic inaccuracies appear correctable, such as a general tendency for SIMON to underestimate scores when the actual IMI score is high.

2) *Significance*: These results show that, while raw LLM-based scoring provides a relatively accurate retrieval of IMI scores, biases that require correction via mathematical transformations can be present in the outputs, in line with related literature [144]–[146]. These findings provide empirical grounding for future methodological development of LLM-based classification and scoring. Specifically, they suggest that such methods should include a process for identifying and correcting statistical biases in raw LLM outputs.

3) *Limitations and future work*: The current work relies on a limited set of LLMs. A broader comparison across a wider variety of LLMs can clarify whether similar biases consistently occur. Likewise, alternative prompting strategies should be considered, as they may reduce or alter these biases. Further work should also evaluate a range of transformation methods to identify possible performance gains.

C. Research Question 3: How does using smaller models and/or in-house LLMs (enabling additional data control and privacy protection) impact the performance of LLM-based IMI scoring?

1) *Insights*: Results show that the highest performance is achieved when the most computationally expensive LLMs are used. Results also show that the degradation in scoring performance remains limited (< 10%) when relying on less computationally expensive models. LLM-based assessment of intrinsic motivation using locally hosted models is technically feasible.

2) *Significance*: These findings demonstrate the applicability of LLM-based scoring of intrinsic motivation across a broad array of contexts. Demonstrating the technical validity of using smaller, faster, and less expensive models provides a clear pathway to cost-efficient and scalable recognition of intrinsic motivation. Demonstrating the viability of locally hosted IMI scoring from natural language also supports compliance with high privacy and data protection standards and regulations such as GDPR. This alleviates barriers that would otherwise make LLM-based scoring of intrinsic motivation from text impracticable in therapeutic contexts, where collected data may be labeled as sensitive personal information and thus highly regulated. In particular, non-local deployments may require manual anonymization of all textual data, which can defeat the purpose of LLM-based scoring (too expensive, too slow) and can introduce legal risks in case of anonymization errors.

From scientific and therapeutic standpoints, these findings enable the practical viability of extracting levels of intrinsic motivation in real-world settings. Examples include extracting

motivation levels in large datasets for scientific purposes and using regular motivation checks during therapeutic activities. Moreover, the scalability and responsiveness implied by these results support alternative approaches for sensing intrinsic motivation, including interactive experimental designs (e.g., a therapeutic robot adapting to patients' motivation levels [147]–[149]), or secondary use of data generated during therapeutic activities (e.g., patient journals, transcripts of exchanges with therapists).

3) *Limitations and future work*: The current work focused on a limited set of smaller LLMs and locally hosted LLMs. This set should be broadened to identify models that further optimize scoring performance, and to identify a Pareto front of trade-offs between cost efficiency, responsiveness, privacy, and scoring performance. Likewise, LLM optimization methods such as fine-tuning should be considered.

Additionally, non-LLM-based methods should be considered (e.g., BERT-based scoring [82], [150], [151]). While such methods typically require large-scale datasets, they can offer relevant trade-offs compared with LLM-based solutions. Their development can also benefit from distillation, that is, using the LLM-based solutions described in this paper as a means of substituting human coding [152]–[154].

D. Research Question 4: How do different methods for eliciting natural language expressions of intrinsic motivation impact the performance of LLM-based IMI scoring?

1) *Insights*: MAE results show that more exhaustive natural language expressions of intrinsic motivation yield the highest scoring performance. Results also show that when reducing the input from seven questions (one general question about motivation and six questions covering the six subscales) to a single question covering the subscale to assess, the loss in scoring performance remains limited ($\leq 10\%$). When reducing the input to only a general question about motivation, a substantial degradation in performance is observed. These results are further supported by the qualitative analysis, which shows that stronger overinterpretations occur when the input text lacks directly relevant indicators, and that such cases were debated by the annotator.

2) *Significance*: These insights are important for future methodological development and practical deployment. In line with prior observations [58], [60], [155], they suggest that good scoring performance may be achieved with relatively limited input data (similarly to [81]), but that data collection should be designed to elicit explicit indicators for each subscale, and implicit cues should be avoided where possible. For example, a therapeutic chatbot should ensure that all subscales are covered and include mechanisms to detect whether each subscale has received a sufficiently explicit answer and, if not, to ask targeted follow-up questions.

3) *Limitations and future work*: The results gathered in this paper are based on a limited set of elicitation methods (seven questions, each answered with at most a few sentences). A wider range of data collection approaches should be considered, such as single-word or short-phrase elicitation (similarly to [81]), longer-form input (e.g., essays), and interactive elicitation (e.g., interviews). Likewise, the data collection context

was limited to online form filling. Future work can assess generalizability by considering different interaction contexts.

E. Risks and Unintended Consequences

Despite the attractive prospects regarding IMI scoring from natural language motivational input brought about by this paper, and the potential benefits for mental health research and practice, the integration of such solutions is often acknowledged to expose unintended implications that may reduce the value provided by these systems, as listed by [156]–[158]. These include challenges regarding adoption (by, e.g., patients, therapists, and social services) [159]–[161], overcritical or undercritical attitudes toward AI-generated content [162], dehumanizing effects in mental health contexts [163]–[165], missing the wider context tied to mental health (e.g., social, cultural, economic) [166], [167], transferring and enforcing implicit norms and values of dominant groups over others (e.g., reduced care for underrepresented populations, stigmatization of neurodivergence) [168]–[170], harms from algorithmic bias [171]–[174], limitations in transparency and accountability, data privacy and security risks [175]–[177], overreliance and dependence on AI systems and their providers [178], [179], and environmental impact [180], [181].

From an interaction perspective, the prospects raised by this paper can be seen as an alternative way of eliciting intrinsic motivation, shifting from close-ended forms or interviews to open-ended expressions of experience, where each elicitation format comes with its own benefits and shortcomings. For example, close-ended forms have the downside of imposing a conceptual framing of intrinsic motivation onto respondents in a highly segmented manner, which may be perceived as colder and less intuitive than asking about one's experience of motivation. However, these forms are effective in requiring respondents to consider all aspects of intrinsic motivation within the underlying framework. Different elicitation methods may also yield differentiated effects across populations. For example, open-ended formats rely more heavily on respondents' capacity to convey their mental state than a form, which may increase the demands on sustained attention.

VII. CONCLUSION

This paper develops an LLM-based method for inferring quantitative measures of intrinsic motivation, as framed by the Intrinsic Motivation Inventory (IMI) [94], [95], [95]–[102], based on open-ended natural language responses to questions about motivation.

The paper makes two main contributions. First, it introduces a database, which we call IMIT (Intrinsic Motivation Inventory and Text), that links, for each participant: (a) an open-ended natural language answer to a general question about motivation; (b) for each of the six IMI subscales, a quantitative subscale score captured by the standard IMI form; and (c) for each of the six IMI subscales, an open-ended natural language answer to a subscale-specific question. This database is made publicly available as supplementary material for this article. It provides a concrete starting point for the affective computing

community to replicate our results and expand research on inferring motivation levels from natural language.

Second, this paper investigates whether IMI scores can be inferred from natural language answers to motivation-related questions, and whether LLM-based solutions can reliably perform this retrieval. We develop twelve implementations of an LLM-based scoring system, which we call SIMON (Scoring Intrinsic MOTivation inventory scores based on Natural language), using prompt engineering [182]–[184]. These variants rely on similar prompt structures, but differ along three core variables: the computational cost of the underlying LLM, whether the LLM can provide stronger privacy features through local hosting, and what type of natural language input is provided (the answers to all seven questions, the answer to the single question pertaining to the IMI subscale to assess, or the answer to the general question about motivation). These variables cover core concerns tied to deploying LLMs in research and practical settings, where scoring performance must be traded off against privacy, costs, and responsiveness.

The twelve variants of SIMON were validated against the IMIT database. Results demonstrate the feasibility of LLM-based solutions for retrieving IMI scores from open-ended answers (Research Question 1). The raw IMI scores retrieved by the top-performing SIMON variant are relatively close to participants' self-reported IMI scores, with an average error of less than one point (i.e., below 15%) (Research Question 2). Further calibration is likely to improve performance, as medium to strong correlations between actual and inferred scores were identified across subscales, and inferred scores showed a tendency to underestimate motivation. In addition, results show that more computationally expensive models and more extensive open-ended inputs lead to the highest scoring performance. Selecting less powerful models, including locally hosted models with stronger privacy features, has only limited impact on scoring performance ($\leq 10\%$) (Research Question 3). A similar loss of performance is observed when the system only accesses the natural language input corresponding to the subscale being measured, whereas a substantial degradation occurs when SIMON is provided only a general motivation text (Research Question 4).

In summary, this study shows that open-ended questions about intrinsic motivation can contain sufficient epistemic material to infer motivation levels, at least as captured by the IMI, and that this inference can be effectively performed by LLM-based solutions. These solutions can meet practical requirements for research and clinical deployment, including relatively low cost, strong data security requirements such as GDPR compliance, and limited respondent burden with minimal degradation in scoring performance [63], [185], [186]. While these results require confirmation and extension prior to clinical deployment, for example by varying participant populations, natural language elicitation methods, and model families, the findings highlight strong prospects for the technical and operational viability of LLM-based motivation recognition in affective computing and mental healthcare. This evidence establishes a necessary first step toward scaling these methods to settings involving clinical populations [187]–[189]. The initial promise underlined by this paper points to high-impact

prospects for mental healthcare by enabling the acquisition and maintenance over time of intrinsic motivation, one of the most central variables acting as a risk factor and symptom for psychological disorders, as well as a primary resource to manage for achieving therapeutic success.

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APPENDIX

You act as a therapist in charge of assessing the intrinsic motivation of a respondent along a predefined dimension.

*Specifically, your objective consists in assessing how true is the respondent displaying *[PROMPT NAME]* regarding their primary activity. This dimension is based on the definition provided by the Intrinsic Motivation Inventory (IMI) described in <https://selfdeterminationtheory.org/intrinsic-motivation-inventory/> and related articles such as:*

*Ryan, R. M., Mims, V., & Koestner, R. (1983). Relation of reward contingency and interpersonal context to intrinsic motivation: A review and test using cognitive evaluation theory. *Journal of personality and Social Psychology*, 45(4), 736.*

*Ryan, R. M. (1982). Control and information in the intrapersonal sphere: An extension of cognitive evaluation theory. *Journal of personality and social psychology*, 43(3), 450.*

The output is formatted as a json output with the following categories:

- "explanation", where you explain the basis of your assessment (max 200 words). Use quotations from the respondent whenever possible.

- "decision", stating the score (a value from 1 to 7, where 1 means not true at all, 4 means somewhat true, and 7 means very true)

The respondent was asked the following question: "[QUESTION ASKED]"

He/she answered the following: "[PROVIDED ANSWER]"

A. Participant input

1) *General motivation:* I get paid to go to work. I also gain robust community connections because I have helped the same people at work for 18 years.

2) *Interest/enjoyment:* I enjoy helping people solve problems. I also enjoy seeing someone and following up with how their life is going from the last time I saw them.

3) *Perceived competence:* I feel very competent because I have been at this job for 18 years. I am able to solve most problems or find help if I get stuck.

4) *Effort/importance:* I sometimes invest too much energy because I need down time to recover. I am an introvert so a lot of social interaction requires me to recharge my battery alone at home.

5) *Pressure/tension:* Sometimes things can get tense such as when customers complain about all the shoplifting. Since my community has voted to slash police budgets and cut the police force, the crime has gone up a lot and it is very hard to keep items in stock. I feel discouraged because even though we often are able to get people arrested, the local judges release people immediately and do not give prison time for shoplifting ever. I do not give up because I like my job otherwise and feel a sense of gratitude for the people I serve.

6) *Perceived choice:* It was a utility choice because I have nervous system problems that do not allow me to work full time on the computer. Some of this is due to having been taught to be a vegetarian while in college. I am just now

solving my nerve problems by eating beef again so maybe I will be able to take a computer job soon. So, I work at my current job because it is one of the few jobs that still exist where I do not have to work on a computer.

7) *Value/usefulness:* I maintain my health and I maintain my connections to the community. I get to move around a lot which is good for my health. I also get out of the house, which is good for my social well being.

8) *Self-reported score:* 3.5

B. Justification of SIMON-gpt5.2-exhaustive on the "effort/importance" subscale

"The respondent reports investing substantial energy and persisting despite challenges, suggesting high effort. They explicitly note, "I sometimes invest too much energy because I need down time to recover," which indicates strong exertion and engagement. Even when the work environment becomes difficult ("Sometimes things can get tense... I feel discouraged"), they emphasize persistence: "I do not give up because I like my job otherwise and feel a sense of gratitude for the people I serve." Their long tenure and problem-solving orientation ("I have been at this job for 18 years... able to solve most problems or find help") further supports sustained effort over time. However, their effort is moderated by introversion and need for recovery ("a lot of social interaction requires me to recharge"), implying effort is high but not consistently maximal across all situations."

1) *SIMON-gpt5.2-exhaustive score:* 6

2) *Human annotator score:* 5